

MongoDB CRUD Operations — Screenshot 1

```
mongosh mongo://127.0.0.1:27017/
Microsoft Windows [Version 10.0.26289.9457]
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C:\Users\dell>mongosh
Current Mongosh Log ID: 6ac18851d5e775868d145924
Connecting to:  mongo://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.12.0
Using MongoDB:  7.0.5
Using Mongosh:  2.12.0

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2026-10-02T14:20:53.585-05:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> show dbs
admin      48.00 KiB
car        72.00 KiB
config     84.00 KiB
local      72.00 KiB
myDatabase 72.00 KiB
studentDb  72.00 KiB
test> use studentDb
switched to db studentDb
studentDb> db
```

MongoDB CRUD Operations — Screenshot 2

```
myDatabase 72.00 KiB
studentDb 72.00 KiB
test> use studentsDB
switched to db studentsDB
studentsDB> db
studentsDB
studentsDB> show collections
```

MongoDB CRUD Operations — Screenshot 3

```
studentsDB> db.students.insertOne({
|   name: "Harini",
|   age: 22,
|   course: "MCA",
|   status: "Active"
| })
{
  acknowledged: true,
  insertedId: ObjectId('6ac18101d5e775860d145925')
}
```

MongoDB CRUD Operations — Screenshot 4

```
studentsDB> db.students.insertMany([
  {
    name: "Priya",
    age: 21,
    course: "BCA",
    status: "Active"
  },
  {
    name: "Kavin",
    age: 23,
    course: "MCA",
    status: "Inactive"
  },
  {
    name: "Arun",
    age: 20,
    course: "BCA",
    status: "Active"
  }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ac18135d5e775860d145926'),
    '1': ObjectId('6ac18135d5e775860d145927'),
    '2': ObjectId('6ac18135d5e775860d145928')
  }
}
```

MongoDB CRUD Operations — Screenshot 5

```
studentsDB> db.students.find()
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 22,
    course: 'MCA',
    status: 'Active'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145928'),
    name: 'Arun',
    age: 20,
    course: 'BCA',
    status: 'Active'
  }
]
studentsDB> db.students.find({ name: "Harini" })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 22,
    course: 'MCA',
    status: 'Active'
  }
]
```

MongoDB CRUD Operations — Screenshot 6

```
]
studentsDB> db.students.findOne({ name: "Harini" })
{
  _id: ObjectId('6ac18101d5e775860d145925'),
  name: 'Harini',
  age: 22,
  course: 'MCA',
  status: 'Active'
}
studentsDB> db.students.countDocuments()
4
studentsDB> db.students.countDocuments({ course: "MCA" })
2
studentsDB> db.students.updateOne(
|   { name: "Harini" },
|   { $set: { status: "Completed" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentsDB> db.students.find({ name: "Harini" })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 22,
    course: 'MCA',
    status: 'Completed'
  }
]
```

MongoDB CRUD Operations — Screenshot 7

```
studentsDB> db.students.updateOne(
|   { name: "Harini" },
|   { $set: { status: "Completed" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentsDB> db.students.find({ name: "Harini" })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 22,
    course: 'MCA',
    status: 'Completed'
  }
]
studentsDB> db.students.updateMany(
|   { course: "BCA" },
|   { $set: { department: "Computer Applications" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
studentsDB> db.students.find({ course: "BCA" })
[
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  },
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 22,
    course: 'BCA',
    status: 'Completed',
    department: 'Computer Applications'
  }
]
```

MongoDB CRUD Operations — Screenshot 8

```
}
studentsDB> db.students.find({ course: "BCA" })
[
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145928'),
    name: 'Arun',
    age: 20,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  }
]
studentsDB> db.students.updateOne(
|   { name: "Harini" },
|   { $inc: { age: 1 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

MongoDB CRUD Operations — Screenshot 9

```
studentsDB> db.students.deleteOne({ name: "Arun" })
{ acknowledged: true, deletedCount: 1 }
studentsDB> db.students.find()
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
```

MongoDB CRUD Operations — Screenshot 10

```
]
studentsDB> db.students.find({ age: { $eq: 23 } })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
studentsDB> db.students.find({ status: { $ne: "Active" } })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
studentsDB> db.students.find({ age: { $gt: 21 } })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {

```

MongoDB CRUD Operations — Screenshot 11

```
studentsDB> db.students.find({ age: { $gte: 21 } })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
studentsDB> db.students.find({ age: { $lt: 23 } })
[
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  }
]
studentsDB> db.students.find({ age: { $lte: 23 } })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
```

MongoDB CRUD Operations — Screenshot 12

```
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
studentsDB> db.students.find({
|   course: { $in: ["BCA", "MCA"] }
| })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
```

MongoDB CRUD Operations — Screenshot 13

```
]
studentsDB> db.students.find({
|   course: { $nin: ["BCA", "MCA"] }
| })

studentsDB> db.students.find({
|   $and: [
|     { age: { $gt: 20 } },
|     { course: "MCA" }
|   ]
| })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
```

MongoDB CRUD Operations — Screenshot 14

```
studentsDB> db.students.find({
|   $or: [
|     { course: "MCA" },
|     { age: 20 }
|   ]
| })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
studentsDB> db.students.find({
|   name: { $not: { $eq: "Harini" } }
| })
[
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
```

MongoDB CRUD Operations — Screenshot 15

```
]
studentsDB> db.students.find({
|   name: { $exists: true }
| })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    status: 'Active',
    department: 'Computer Applications'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    status: 'Inactive'
  }
]
```

MongoDB CRUD Operations — Screenshot 16

```
]
studentsDB> db.students.updateMany(
|   {},
|   { $rename: { status: "studentStatus" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
studentsDB> db.students.find()
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    studentStatus: 'Completed'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    department: 'Computer Applications',
    studentStatus: 'Active'
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    studentStatus: 'Inactive'
  }
]
```

MongoDB CRUD Operations — Screenshot 17

```
studentsDB> db.students.updateOne(
|   { name: "Harini" },
|   { $unset: { department: "" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
studentsDB> db.students.updateMany(
|   {},
|   { $push: { skills: "Java" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

MongoDB CRUD Operations — Screenshot 18

```
studentsDB> db.students.find()
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    studentStatus: 'Completed',
    skills: [ 'Java' ]
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    department: 'Computer Applications',
    studentStatus: 'Active',
    skills: [ 'Java' ]
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    studentStatus: 'Inactive',
    skills: [ 'Java' ]
  }
]
```

MongoDB CRUD Operations — Screenshot 19

```
studentsDB> db.students.updateMany(
|   {},
|   { $pull: { skills: "Java" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
studentsDB> db.students.find().sort({ name: 1 })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    studentStatus: 'Completed',
    skills: []
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    studentStatus: 'Inactive',
    skills: []
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    department: 'Computer Applications',
    studentStatus: 'Active',
    skills: []
  }
]
```

MongoDB CRUD Operations — Screenshot 20

```
]
studentsDB> db.students.find().sort({ age: -1 })
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    studentStatus: 'Completed',
    skills: []
  },
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    studentStatus: 'Inactive',
    skills: []
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    department: 'Computer Applications',
    studentStatus: 'Active',
    skills: []
  }
]
```

MongoDB CRUD Operations — Screenshot 21

```
]
studentsDB> db.students.find().limit(2)
[
  {
    _id: ObjectId('6ac18101d5e775860d145925'),
    name: 'Harini',
    age: 23,
    course: 'MCA',
    studentStatus: 'Completed',
    skills: []
  },
  {
    _id: ObjectId('6ac18135d5e775860d145926'),
    name: 'Priya',
    age: 21,
    course: 'BCA',
    department: 'Computer Applications',
    studentStatus: 'Active',
    skills: []
  }
]
```

MongoDB CRUD Operations — Screenshot 22

```
studentsDB> db.students.find().skip(2)
[
  {
    _id: ObjectId('6ac18135d5e775860d145927'),
    name: 'Kavin',
    age: 23,
    course: 'MCA',
    studentStatus: 'Inactive',
    skills: []
  }
]
```